

One time pad

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One time pad

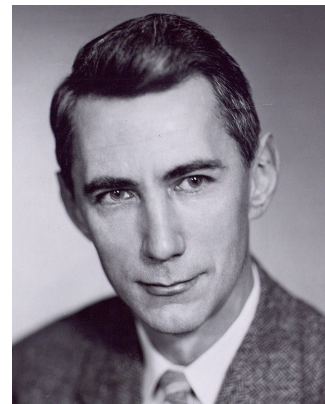
- An encryption technique which cannot be cracked.
- Requires a pre-shared key
 - Larger than or equal to the size of the message
- Each bit of character of the plaintext is encrypted by combining it with the corresponding bit or character from the pad.
 - By using modular addition.
 - XOR.

2 input XOR gate

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Secrecy

- The ciphertext must not give any hints or clues about the plaintext.
 - If there is such a scheme, it is called **perfect secrecy**.
 - This term is coined by Claude Shannon in 1949.
- In OTP, the key is **very long**. Therefore, sometimes people think using an excerpt from the book etc.
 - We cannot talk about *randomness* and *perfect secrecy* anymore.
 - It is **not** a secret if taken from a book!
- The key in one time pad **must not be reused**.
 - **ONE** time pad.



Perfect Secrecy

- Ciphertext should provide no information about plaintext.
 - Such a scheme has **perfect secrecy**.
- One time pad has **perfect secrecy**. Therefore it cannot be cracked.
- Term is coined by Claude Shannon in 1949.
- Perfect secrecy requires :
 - $| \text{Key Space} | \geq | \text{Message Space} |$
 - This is why OTP key must be as long as the message.

One Time Pad

- Brute force does not help.
- If you try all the keys, you will just produce **every possible message** including the **wrong ones**
- Example
- If we try all keys on: NHYYSLKYBMNJA
 - We will get **every 13 letter string**.
 - Including “MEET ME OUTSIDE” and “MEET ME INSIDE”. The attacker cannot tell which is right.

One-time pad

- If the following conditions are met, one time pad cannot be **cracked**.
 - Key must be as long as the plaintext.
 - Key must be truly random.
 - Key must never be reused.
 - Key must be kept completely secret.
- If even one of these are not met, it is **not** a one time pad.
- The security is not based on the computational hardship (we'll see that in pke), but in the randomness of the key.
- Perfectly secret because for any plaintext and ciphertext, **exactly one key maps between them**.

Difficult to use in practice

- Hard to generate truly random, long keys. (**key generation**)
 - What if you want to encrypt 100 MB of data?
 - What would be the key size?
 - $100\text{MB} = 100,000,000 \text{ bytes} \sim 800,000,000 \text{ bits}$
- Key Distribution
- Key Storage
- ..

Ho

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

One-Time Pad Encryption

MEET ME OUTSIDE

B D U F G H W E I U F G W
D L K N F L N D K L F N L K
I R E U P O W Q I R P N M A
J C M L W O I D Y C H N S J
V B X N L Z O W U E O R P
N S J S K A K E O I R Y W I S

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Plaintext:	M	E	E	T	M	E	O	U	T	S	I	D	E
Numerical Plaintext:	12	4	4	19	12	4	14	20	19	18	8	3	4
OTP:	B	D	U	F	G	H	W	E	I	U	F	G	W
Numerical OTP:	1	3	20	5	6	7	22	4	8	20	5	6	22
Numerical Ciphertext:	13	7	24	24	18	11	10	24	1	12	13	9	0
Ciphertext:	N	H	Y	Y	S	L	K	Y	B	M	N	J	A

Therefore the ciphertext is "NHYYSLKYBMNJA".

How it works?

- You can also use the ASCII representation and use XOR.
-

ENCRYPT		
$\oplus$	0 0 1 1 0 1 0 1	Plaintext
	1 1 1 0 0 0 1 1	Secret Key
=	1 1 0 1 0 1 1 0	Ciphertext
DECRYPT		
$\oplus$	1 1 0 1 0 1 1 0	Ciphertext
	1 1 1 0 0 0 1 1	Secret Key
=	0 0 1 1 0 1 0 1	Plaintext

Never reuse the key

- If we same key **k** encrypts two messages:
 - $C1 = P1 \text{ XOR } k$
 - $C2 = P2 \text{ XOR } k$
- If we XOR C1 and C2, it will be the same with P1 XOR P2.
- So now, the attacker never saw the key, but holds the XOR of two plaintexts.
- From there, by using language statistics and guessable words (*the, attack*, file headers), they can recover both messages.
- With XOR only ciphers, **reuse is catastrophic**

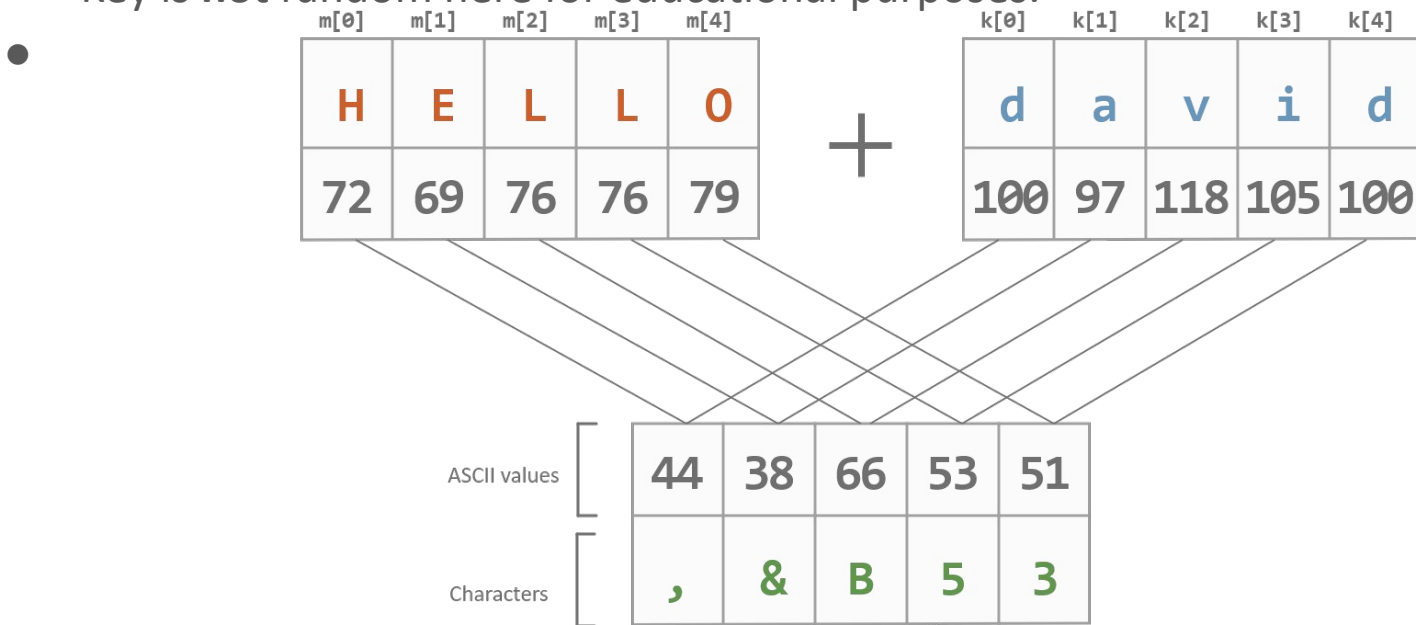
Malleability

- OTP hides the message but does not protect it from modification.
- An attacker who knows or guesses the plaintext format can change it without knowing the key.
- Imagine that there is a plaintext: PAY \$100.
 - If the attacker **flips** a single bit, everything can change.
- 1 = 00110001
- 9 = 00111001
- The only difference between these two numbers are a **single bit**. So if you flip that, the meaning will totally change.
- It will become PAY \$900.

- For that reason, real systems add a MAC (we'll see that later) or use authenticated encryption.
 - AE is a method that simultaneously provides data confidentiality, data integrity and message authenticity.
- It leaks **message length**.
 - This can be prevented by using *padding*.
- But does not give anything about the content.

How it works?

- Or you can even use ASCII but decimal representation.
- Don't forget mod 128.
- Key is **not** random here for educational purposes.



Real life usage

- Originally described by a banker in USA in 1882.
 - Frank Miller
- It was reinvented in 1917 by Gilbert Vernam and Joseph Mauborgne.
 - Vernam Cipher (a specific implementation of OTP)
 - Not every Vernam cipher is necessarily an OTP.
- OTP was used by different countries.
 - Consists of stack of small very thin pages, each with a series of random 5-digit numbers.
 - Each page is destroyed after use.



Real life usage (Distribution)

- A unique set of booklets needs to be issued to each agent.
- Since you destroy when you use them, you need a sufficient supply.
- East-german secret service used a travel kit.



courtesy of avid netherlands

Real life usage (concealment)

- One other way is to print them very small and hiding them inside common objects.
 - Ballpoints, photoframes, walnut.
 - Popular with KGB agents in Western Europe.



courtesy of cryptomuseum

One time pad (variations)

- Some OTP contains only numbers.
 - If this is the case, you need to convert your plaintext into numbers.
- Some of them contains letters.

Stream Ciphers

- OTP gives us perfect secrecy, but the key must be **as long as the message**, must be random and never reused.
- What if we are able to stretch a **short secret seed** into a long **random-looking** keystream?
 - Yes, we will give up **perfect secrecy** and we will settle for **computational security**.

References

- <https://www.ciphermachinesandcryptology.com/en/onetimepad.htm>
- <https://www.cryptomuseum.com/crypto/otp/index.htm>